



Assignment: Ad-Enabled LLM as a Service

Course Project (Topic 8)

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Abstract

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1 Introduction

1.1 Background and Motivation

The dominant paradigm in digital advertising has, for two decades, relied on a fundamental assumption: the user interacts with a visual interface—a web page, a search results listing, a social media feed—where rectangular display regions can be allocated to sponsored content. Banner advertisements, programmatic display auctions, and search-engine sponsored links all exploit this property. The advertiser purchases screen real estate; the publisher renders a creative unit; the user either attends to it or does not. This model, refined through successive generations of behavioural targeting and real-time bidding, generates hundreds of billions of dollars in annual revenue worldwide and underpins the economic viability of most free-to-use internet services.

Conversational AI interfaces disrupt this assumption at its foundation. Systems such as ChatGPT, Claude, and Gemini present information as a single stream of natural-language text. There are no sidebars, no interstitial slots, and no visual regions in which a banner can be placed without breaking the conversational contract between the system and its user. At the same time, these interfaces are absorbing an increasing share of the queries that users previously directed to search engines—product research, comparison shopping, recommendation requests—precisely the query types that carry the highest commercial intent and, consequently, the highest advertising value. The shift is not merely a change of medium; it is a structural transformation in how users express intent, receive information, and make purchasing decisions.

The advertising industry therefore faces a concrete technical problem: how to monetise conversational AI interactions without degrading the user experience that makes them valuable in the first place. Appending a block of sponsored links to the end of a chat response replicates the search-ad format but sacrifices the natural feel of the conversation. Inserting interstitial advertisements between turns interrupts the dialogue flow and erodes user trust. Neither approach exploits the distinctive capability of a large language model (LLM), namely its ability to generate contextually appropriate, fluent natural-language text that integrates information from multiple sources—including sponsored product information—into a single coherent response.

This observation motivates the present work. If the LLM can understand the user’s intent, match it against an inventory of relevant products, and weave a sponsored recommendation naturally into an otherwise helpful response, then the advertisement ceases to be an interruption and becomes, instead, part of the answer. The result is an advertising format that is native to the conversational medium: contextually relevant, linguistically integrated, and measurable not merely through impressions and clicks but through the user’s subsequent conversational engagement with the recommended product.

The same logic extends beyond text. Conversational AI interfaces increasingly support image generation as a native modality, and a user who asks the system to render a scene is expressing creative and often commercial intent in a form that is no less monetisable than a textual product query. If the language model can weave a sponsored product into a written answer, an image model can, by analogous reasoning, place that product naturally within a generated scene—on a shelf, in a character’s hand, within the environment the user described—rather than overlaying a banner on top of the artwork. Treating text and image generation as two surfaces of a single advertising pipeline is therefore a natural extension of the core thesis, and one that this project

pursues explicitly.

A workable platform, however, cannot consist of the generative pipeline alone. A production-grade advertising system also requires the operational scaffolding that makes it usable by real advertisers: a way to onboard them, a way to organise their spend into campaigns with start and end dates and daily pacing, a way to create and manage ads without touching the database directly, and a way for both the platform operator and the advertiser to observe how their money is being spent. These concerns are conventionally relegated to “future work,” but they fundamentally shape the data model and the request path—budget gating, for example, must happen inside the ad-retrieval loop, not as an afterthought—and this project therefore treats them as first-class deliverables rather than deferred extensions.

1.2 Project Objectives

This project designs, implements, and evaluates an end-to-end LLM-powered conversational advertising platform—referred to throughout this report as *Ad LLM as a Service*—that delivers inline sponsored product recommendations within AI chat responses. The platform satisfies the following objectives:

- (i) **Intent-aware context extraction.** A lightweight LLM call extracts structured context from each user message and the preceding conversation history, producing a classification of intent, topical relevance, sentiment, purchase signal strength, and—critically—a sensitivity-aware ad receptivity score that gates whether any advertisement should be shown at all.
- (ii) **Semantic ad matching with budget-aware eligibility.** The extracted context is embedded into a vector representation and matched against a pre-embedded ad inventory via cosine similarity search over `pgvector`. Candidate ads are then filtered through a campaign eligibility check—verifying that the parent campaign is active, within its scheduled window, and has not exhausted either its total budget or its daily cap—and re-ranked by a business-rule stage that incorporates bid price, per-session frequency caps, and brand-safety tags.
- (iii) **Natural ad integration with mandatory disclosure.** The main response-generation LLM receives the matched advertisement as part of its prompt context and is instructed to weave the product mention into a helpful answer if and only if the product is genuinely relevant. Every sponsored mention carries a `[Sponsored]` disclosure tag, ensuring compliance with Federal Trade Commission (FTC) endorsement guidelines.
- (iv) **Multi-dimensional response and engagement evaluation.** Beyond conventional impression and click-through tracking, a dedicated LLM-as-judge module analyses each post-ad follow-up message and returns a structured set of metrics: an engagement type classification (product inquiry, comparison, price inquiry, purchase intent, feature question, positive or negative reaction, dismissal), a continuous engagement score, sentiment toward the advertisement, a naturalness score capturing how intrusive the placement felt, and an estimated purchase-proximity score. These metrics jointly form a higher-fidelity signal than any single click-or-not-click measurement and are the substrate on which downstream optimisation operates.
- (v) **Ethical safeguards.** Conversations classified as sensitive—those involving health crises,

emotional distress, financial hardship, or grief—are automatically excluded from ad injection, regardless of commercial relevance.

- (vi) **Adaptive per-ad context optimisation.** Each advertisement in the inventory carries a dynamic presentation-guidance document—its *ad context*—which is injected into the response-synthesis prompt alongside the product description. An initial version of this context is generated automatically by an LLM at the moment the ad is created, so that every ad begins life with a coherent presentation strategy rather than an empty field. A background optimiser then periodically reviews the accumulated engagement metrics for each ad, diagnoses what is working and what is not (low naturalness, high dismissal rate, weak purchase proximity despite strong topical match), and asks a reasoning LLM to rewrite the context guide accordingly. Every revision is versioned and persisted to an `ad_context_history` table together with the metrics snapshot that triggered it and a natural-language justification, yielding a fully auditable record of how each ad’s presentation strategy evolved over time.
- (vii) **Image generation with organic product placement.** The platform exposes a second generative surface alongside text chat: a user-prompted image-generation endpoint that routes through the same context extraction and ad matching pipeline. When a relevant sponsored product is identified, the user’s original image prompt is rewritten by an auxiliary LLM into a placement prompt that fits the product into the requested scene, and the final image is rendered by a diffusion model (`gpt-image-1`). When the advertised product has a reference photograph, it is supplied to the image model via an edit-mode call so that the rendered product is visually grounded in the real article rather than hallucinated from its textual description.
- (viii) **Advertiser portal with campaign and budget management.** A self-serve REST API and accompanying React interface allow advertisers to register, organise their spend into campaigns with optional start dates, end dates, and daily budget caps, and create ads within each campaign. Campaigns follow an explicit lifecycle—**draft**, **active**, **paused**, **completed**—with auto-completion when the total budget is exhausted, and daily spend is reset lazily at first touch each new day rather than via a cron job. Ad creation is accelerated by a URL-based extraction flow in which the advertiser pastes a product page URL, the server scrapes it, and an LLM call populates the ad fields (name, description, creative copy, target topics and intents, brand-safety tags) as a pre-filled draft that the advertiser can review and edit before submission.
- (ix) **Dual analytics dashboards.** The platform ships two distinct analytics surfaces backed by a shared event log. A global, platform-wide dashboard reports aggregate impressions, click-through rate, engagement events, estimated revenue, and top-performing ads across all advertisers, intended for the platform operator. An advertiser-facing dashboard reports the same metrics scoped to a single advertiser or a single campaign, with a daily breakdown driven by the `daily_spend_log` table and visual indicators of budget utilisation.
- (x) **Functional MVP.** The deliverable is a working prototype that demonstrates the complete pipeline from user message to ad-augmented response—across both text and image modalities—together with the advertiser portal, the adaptive context optimiser, and both analytics dashboards.

1.3 Scope and Constraints

The platform is scoped as a minimum viable product (MVP) and is subject to several deliberate constraints that bound the design space.

The architecture is a single-service deployment: one FastAPI orchestrator coordinates all internal modules, a single Redis instance manages session state, and a single PostgreSQL database (extended with `pgvector`) stores the ad inventory, the advertiser and campaign records, the daily spend log, the ad-context history, and the event log. This monolithic design is appropriate for the current scale—tens of thousands of ads, moderate request concurrency—and avoids the operational complexity of a distributed microservice architecture.

Two external dependencies impose hard constraints on latency and cost. Every chat request requires at minimum two LLM API calls—one lightweight call for context extraction and one full-model call for response synthesis—plus one embedding API call for the query vector. The total end-to-end latency budget is three seconds, inclusive of all network round-trips and internal processing. The per-request cost is dominated by these API calls, and the re-ranking stage uses heuristic scoring weights (a weighted combination of semantic similarity, normalised bid price, and purchase signal) rather than a learned model, since no click-through data is available at launch. Image generation is treated as a separate interactive surface with its own, looser latency budget, since diffusion-model calls routinely exceed ten seconds and are dominated by GPU inference time rather than by the orchestration overhead that constrains the text path.

The engagement evaluator is implemented as a single LLM call that returns the structured metrics described in Section 1.2. It runs asynchronously, off the critical request path, so its latency does not count against the three-second response budget; the trade-off is that engagement metrics become available only after a short delay rather than synchronously with the turn in which the ad was shown.

The adaptive context optimiser is similarly a deferred, out-of-band process. Optimisation for a given advertisement is gated by two guards—a minimum of five recorded engagements and a minimum interval of twenty-four hours since the previous revision—which together prevent the system from overfitting to noise and from thrashing the ad context under light traffic. An advertisement with insufficient traffic therefore retains its LLM-generated initial context indefinitely, and optimisation only begins once the statistical foundation is adequate.

The advertiser portal is scoped to the operational primitives needed to run a campaign end-to-end: advertiser registration, campaign CRUD with budget and schedule controls, ad CRUD with URL-based pre-filling, and scoped analytics. Authentication middleware, invoicing, payment processing, and advertiser self-service account recovery are deliberately out of scope for the MVP; the portal assumes a trusted operator environment and defers commercial-grade identity and billing to subsequent iterations.

1.4 Report Organisation

The remainder of this report is organised as follows. Section 2 surveys the relevant literature across four areas: the evolution of digital advertising formats, the capabilities of large language models for controlled generation, retrieval-augmented generation as a paradigm for context-aware information retrieval, and the emerging body of work on advertising within conversational

interfaces. Section 3 formulates the problem precisely, identifying the technical challenges of real-time intent extraction, low-latency semantic ad matching, natural ad integration, engagement measurement, and per-campaign budget enforcement, together with the ethical and regulatory constraints that any solution must satisfy. Section 4 presents the system architecture and the design of each component: the FastAPI orchestrator, the context extraction module, the Redis-backed session manager, the `pgvector`-based ad matching pipeline with budget-gated eligibility, the prompt-engineered response synthesis module, the image-generation endpoint with product-placement prompt rewriting, the LLM-as-judge engagement evaluator, the adaptive ad-context optimiser, the advertiser portal with its URL-based ad extraction flow, and the dual analytics subsystem serving both the platform-wide and advertiser-scoped dashboards. Section ?? describes the implementation in detail, including workstream responsibilities, the database schema, API contracts, and the quality assurance framework. Section 5 walks through end-to-end examples that exercise representative paths through the pipeline, including successful ad injection, sensitive-conversation suppression, irrelevant-ad rejection, engagement follow-up detection, multi-turn context accumulation, an image-generation request with organic product placement, a campaign reaching its daily budget cap, and an advertiser creating a new ad via URL extraction. Section 6 summarises the contributions, discusses limitations, and outlines directions for future work.

2 Literature Review

This section surveys the bodies of work that inform the design of the *Ad LLM as a Service* platform. The review is organised by *approach* rather than by chronology: each subsection isolates a distinct technical or methodological strand—digital advertising formats, large language model capabilities, retrieval-augmented generation, conversational-interface monetisation, prompt engineering for controlled output, and engagement measurement—and positions the present project relative to the state of the art in that strand. Section 2.9 synthesises the findings and identifies the specific research gap that this project addresses.

2.1 Evolution of Digital Advertising

The economic architecture of online advertising has undergone three major structural transitions since the first banner advertisement appeared on HotWired in 1994, an AT&T creative unit that reportedly achieved a click-through rate exceeding 40% [20].

From display to programmatic. The first generation of digital advertising was transactional and manual: publishers sold rectangular display regions at fixed rates per thousand impressions, and advertisers negotiated placements directly with individual sites. The introduction of ad servers—most notably DoubleClick in 1996—centralised delivery and measurement, but the buying process remained fundamentally a media-planning exercise. The emergence of real-time bidding (RTB) between 2007 and 2010 transformed this model into an auction-based marketplace in which individual impressions are priced algorithmically on the basis of predicted user value [47]. Demand-side platforms (DSPs), supply-side platforms (SSPs), and data management platforms (DMPs) together created a programmatic stack that, by 2022, accounted for approximately 84% of global digital display spending [12].

Search advertising and intent-based targeting. Google’s introduction of AdWords in 2000, later renamed Google Ads, established a second paradigm: intent-based targeting. Rather than inferring user interest from page context or behavioural profiles, search advertising matches sponsored results directly to the user’s expressed query. This alignment of commercial supply with declared demand yields conversion rates substantially higher than those of display advertising and has made search the single largest category of digital ad spending. In 2024, Google’s total advertising revenue reached approximately \$265 billion, with the majority derived from search placements [40]. The economic logic is straightforward: a user who searches for “best running shoes for flat feet” has explicitly declared a purchase-adjacent intent that no amount of behavioural inference on a display network can reliably reconstruct.

Social and behavioural targeting. A third strand developed in parallel on social-media platforms, where dense self-reported demographic data and interaction graphs enabled highly granular audience segmentation. Facebook (now Meta), Instagram, and TikTok built advertising businesses on the principle that a user’s social graph, expressed interests, and engagement patterns constitute a rich behavioural profile against which ads can be targeted even in the absence of an explicit query. This model trades the precision of declared intent for the breadth of passive behavioural signal and has proven effective for awareness and consideration campaigns, though it depends on third-party cookies and device identifiers that are now being phased out under privacy regulations such as the General Data Protection Regulation (GDPR) and Apple’s App Tracking Transparency framework [2], [13].

The move toward native formats. Across all three paradigms, a persistent trend has been the migration from visually distinct ad units—banners, interstitials, pop-ups—toward formats that are structurally integrated with the surrounding content. Native advertising, defined as paid content that matches the form and function of the host medium, arose as a response to banner blindness and ad-blocker adoption [45]. Sponsored articles on news sites, promoted posts on social feeds, and influencer endorsements on video platforms all share the property that the advertisement is embedded within the content stream rather than placed adjacent to it. The platform examined in this report extends this principle to its logical terminus in conversational AI: the sponsored recommendation is not merely placed alongside the response but woven into it by the language model itself.

2.2 Large Language Models and Conversational AI

The transformer architecture. The architectural foundation of the present system is the transformer, introduced by Vaswani et al. [42]. The transformer replaces the recurrent processing of earlier sequence-to-sequence models with a self-attention mechanism that computes pairwise relevance scores across all positions in the input simultaneously, enabling parallelised training and effective capture of long-range dependencies. This architecture has become the dominant substrate for language modelling: every major large language model (LLM) deployed as of 2025—including OpenAI’s GPT series [7], [31], Anthropic’s Claude family [1], Google’s Gemini [18], and Meta’s LLaMA [41]—is built on transformer variants. The progression from GPT-2’s 117 million parameters to GPT-4’s estimated hundreds of billions illustrates the scaling trajectory that has yielded the emergent capabilities exploited in this project [44].

Capabilities relevant to this project. Four capabilities of modern LLMs are directly exercised by the Ad LLM platform:

- (i) *Instruction following.* Post-training alignment techniques—reinforcement learning from human feedback (RLHF) [32] and constitutional AI [3]—have produced models that reliably follow detailed natural-language instructions, including multi-constraint prompts that specify output format, tone, and content restrictions.
- (ii) *Structured output generation.* When instructed appropriately, LLMs generate syntactically valid JSON, markdown, or other structured formats with high consistency. This capability underpins the context extraction module, which must return a machine-parseable classification of intent, topics, sentiment, and ad receptivity from each user message.
- (iii) *Contextual reasoning over long inputs.* Context windows of 100 000 tokens and above—available in models such as Claude 3 (200 000 tokens) and Gemini 1.5 (up to 1 000 000 tokens) [1], [18]—allow the model to condition its generation on extensive conversation histories, a prerequisite for maintaining coherent multi-turn advertising integration.
- (iv) *Tone adaptation.* LLMs modulate register, formality, and lexical choice in response to system-prompt instructions, enabling the response synthesis module to match the conversational style of the user while integrating a sponsored product mention.

Lightweight versus capable models. A practical distinction that shapes the system architecture is the trade-off between model capability and inference cost. Lightweight models—exemplified by Anthropic’s Haiku tier—offer low latency and low per-token cost at the expense of reduced reasoning depth. Capable models—exemplified by the Sonnet tier—produce higher-quality, more nuanced text but at greater cost and latency. The Ad LLM platform exploits this spectrum deliberately: context extraction, a classification task requiring structured output but not creative generation, is assigned to a Haiku-class model; response synthesis, which demands natural integration of a product mention into a helpful answer, is assigned to a Sonnet-class model [1]. This two-tier strategy is consistent with the broader industry practice of routing sub-tasks to the smallest model that achieves acceptable quality, thereby minimising aggregate inference cost [10].

2.3 Retrieval-Augmented Generation (RAG)

The RAG paradigm. Retrieval-augmented generation, formalised by Lewis et al. [25], addresses a fundamental limitation of parametric language models: their knowledge is frozen at training time and cannot be updated or extended without retraining. The RAG architecture augments the generative model with a non-parametric memory—a corpus of documents indexed for efficient retrieval—so that, at inference time, the model conditions its output on both the user query and the most relevant retrieved passages. Lewis et al. demonstrated that this combination of a pre-trained seq2seq transformer (BART) with a dense vector index of Wikipedia, accessed via a neural retriever (Dense Passage Retriever, DPR) [23], achieved state-of-the-art results on open-domain question answering benchmarks and generated more factual, specific language than parametric-only baselines.

Vector embeddings and semantic similarity. The retrieval component of a RAG pipeline depends on dense vector representations of both the query and the candidate documents.

Embedding models—such as OpenAI’s `text-embedding-3-small`, Cohere’s `Embed`, and open-source alternatives like BGE and E5 [43], [46]—map text into high-dimensional vector spaces in which semantic similarity corresponds to geometric proximity (typically measured by cosine distance). The quality of these embeddings determines retrieval precision: a query embedding that accurately captures the user’s informational need will surface documents whose content is genuinely relevant, while a poor embedding will introduce noise that degrades the generator’s output.

Vector databases and extensions. At scale, approximate nearest-neighbour (ANN) search over embedding vectors requires specialised indexing. Dedicated vector databases such as Pinecone, Weaviate, and Milvus provide managed infrastructure for this purpose [21]. For applications with moderate inventory sizes—tens of thousands of records rather than billions—the `pgvector` extension to PostgreSQL offers a pragmatic alternative: it adds vector-typed columns and cosine-distance operators to a relational database, avoiding the operational overhead of a separate vector store [24]. The Ad LLM platform adopts `pgvector` for precisely this reason: the ad inventory is expected to contain at most tens of thousands of items, and co-locating vector search with the relational schema for budget management, frequency capping, and event logging simplifies both development and deployment.

Ad matching as a specialised RAG pipeline. The conceptual contribution of the present project’s ad-matching subsystem is the reinterpretation of RAG in a commercial context. In a conventional RAG system, the retrieval step fetches passages that are informationally relevant to the user’s question; in the Ad LLM pipeline, the retrieval step fetches *advertisements* that are semantically relevant to the user’s conversational context. The query embedding is constructed not from the raw user message but from the structured context extracted by the lightweight LLM—topics, intent, and entities—ensuring that the retrieval signal reflects interpreted user need rather than surface lexical overlap. A business-rule re-ranking stage then applies non-semantic constraints (budget availability, bid price, per-session frequency caps) before the top candidates are passed to the generator. This architecture preserves the retrieve-then-generate structure of RAG while specialising both the retrieval objective and the post-retrieval filtering to the requirements of advertising.

2.4 Advertising in Conversational Interfaces

Industry deployments. The integration of advertising into conversational AI is a recent and rapidly evolving development. Microsoft was the first major platform to insert sponsored content into a conversational interface, embedding text ads, shopping campaigns, and multimedia placements into Bing Chat responses shortly after its launch in February 2023 [29]. In September 2023, Microsoft announced “Conversational Ads,” a new format designed specifically for the chat medium, including “Compare & Decide Ads” that present structured product comparisons within the dialogue flow [28]. Google followed with experiments in its Search Generative Experience (SGE), placing sponsored results both adjacent to and within AI-generated answer summaries, labelled with a bold “Sponsored” tag [16]. By 2025, Google had extended these placements into AI Mode, a fully conversational search interface powered by Gemini 2.0, and reported that ads drawn from existing Search and Shopping campaigns were being served within AI-generated responses [17].

These deployments share a common limitation from the perspective of the present project: they append or interleave conventional ad units (text links, product cards) alongside AI-generated text rather than instructing the language model to integrate the product mention *into* the response itself. The result is a hybrid layout in which the advertisement remains visually and linguistically distinct from the organic answer—a format that is arguably native to the search-results page but not to the conversational medium.

Academic work on native advertising and user perception. A substantial body of research examines how users perceive and respond to native advertising. Wojdynski and Evans [45] demonstrated that fewer than one in ten participants recognised article-style native advertising as paid content, even when disclosure labels were present, highlighting the tension between format integration and consumer awareness. Subsequent work has shown that this tension operates through two competing mechanisms [11]: explicit sponsorship disclosures activate persuasion knowledge, prompting critical evaluation and reducing brand attitude, but simultaneously enhance perceived transparency, which increases trust and partially offsets the negative effect. The balance between these mechanisms depends on disclosure language, placement, and prominence [8], [45]. A Nielsen study reported that 79% of users correctly identified content as advertising when the label read “Advertisement,” compared to only 48% for the label “Presented by” [30]. These findings directly inform the design decision in the Ad LLM platform to use a mandatory, visually rendered [Sponsored] tag—a disclosure format that is both explicit and machine-enforceable.

Ethical considerations: transparency and trust. The ethical literature on advertising transparency converges on a central finding: consumers respond more favourably to advertising when they perceive that the advertiser is being honest about the commercial nature of the content [8], [9]. Transparency functions as a signal of brand integrity [22], and the downstream effects on trust and purchase intention are robust across experimental replications. In the conversational AI context, the stakes are amplified: the user has entered a dialogue that they expect to be helpful and impartial, and undisclosed commercial influence would constitute a violation of that implicit contract. The Ad LLM platform addresses this risk through three mechanisms: (i) the [Sponsored] tag is mandatory and enforced at the prompt level; (ii) the response-generation prompt instructs the model to ignore the advertisement if it is not relevant to the user’s question; and (iii) conversations classified as sensitive are automatically excluded from ad injection; and (iv) each ad carries brand-safety tags that allow advertisers to declare topics their creative must not appear alongside, providing a second, advertiser-controlled filter layered on top of the platform’s sensitivity gate.

Regulatory frameworks. The Federal Trade Commission (FTC) revised its Endorsement Guides in July 2023, updating the definition of “endorsement” to encompass digital formats including social-media tags, virtual influencers, and AI-generated recommendations [14]. The revised Guides require that any material connection between an advertiser and an endorser be disclosed “clearly and conspicuously,” using language that is difficult to miss and easily understood by the target audience. The Guides explicitly note that platform-provided disclosure tools (e.g., Instagram’s “Paid partnership” label) may not be sufficient on their own. Although the FTC has not yet issued guidance specific to LLM-generated sponsored content, the general

principle—that consumers must be informed when content is commercially motivated—applies directly. The [Sponsored] disclosure mechanism in the Ad LLM platform is designed to satisfy this principle by ensuring that every sponsored product mention is preceded by an unambiguous, machine-enforced marker.

2.5 Prompt Engineering for Controlled Generation

Techniques for reliable structured output. Prompt engineering—the practice of crafting input instructions to elicit desired model behaviour without modifying model parameters—has emerged as a critical discipline in applied LLM systems. Schulhoff et al. [38] present a taxonomy of 58 distinct prompting techniques, ranging from zero-shot and few-shot prompting to chain-of-thought reasoning, self-refinement, and retrieval-augmented prompting. Sahoo et al. [37] provide a complementary survey organised by application area, demonstrating that prompt design significantly affects output quality across tasks including question answering, code generation, and text classification. For the Ad LLM platform, the most relevant techniques are *role assignment* (instructing the model to adopt a specific persona), *output-format specification* (requiring JSON or markdown structure), and *conditional instruction* (directing the model to include or exclude content based on stated criteria).

System prompts and instruction following. The system prompt—a privileged instruction block that precedes the user’s message and persists across turns—is the primary mechanism through which the Ad LLM platform controls the response-generation model. The system prompt specifies: the model’s role (a helpful assistant), the conditions under which a sponsored product should be mentioned (relevance to the user’s question), the required disclosure format ([Sponsored] tag), and the link format (markdown with tracking URL). Ouyang et al. [32] showed that instruction-tuned models follow system-level directives with high fidelity, though compliance is not absolute: adversarial or ambiguous inputs can cause the model to deviate from instructions. The prompt design for the Ad LLM platform therefore includes explicit rejection logic (“If the product is not relevant to the conversation, ignore it entirely”) and is validated against an evaluation set of 20–30 scenarios covering both successful integration and correct rejection.

Balancing helpfulness with ad integration naturalness. A distinctive prompt-engineering challenge in this project is the dual-objective nature of the generation task: the response must simultaneously be *helpful* (answering the user’s question accurately and completely) and *commercially effective* (weaving a product mention into the answer in a way that feels like an informed suggestion rather than an insertion). These objectives are not inherently conflicting—a relevant product recommendation *is* helpful—but they become so when the matched advertisement is only tangentially related to the user’s query. The prompt must therefore encode a threshold: the model should integrate the product only when doing so genuinely enhances the response, and should produce a purely organic answer otherwise. Achieving consistent behaviour across diverse conversation types requires iterative prompt refinement guided by systematic evaluation.

LLM-as-judge for automated quality evaluation. Manual evaluation of generated responses does not scale. The LLM-as-judge paradigm, surveyed by Zheng et al. [48] and Gu

et al. [19], addresses this limitation by using a separate LLM call to score generated outputs along specified quality dimensions. Zheng et al. demonstrated that GPT-4, when used as a judge, achieves approximately 80% agreement with human preference scores—matching the inter-annotator agreement rate among human raters themselves. The Ad LLM platform adopts this paradigm for offline quality monitoring: a judge model scores sampled production responses on helpfulness, naturalness of ad integration, disclosure accuracy, and ad relevance, enabling automated detection of quality regressions without requiring continuous human review.

Beyond sampled offline monitoring, the Ad LLM platform extends the LLM-as-judge paradigm into an online measurement role: every follow-up turn after an ad impression is passed to a judge model that returns a structured multi-dimensional assessment comprising an engagement score, an ad-naturalness score, an estimated purchase-proximity score, a sentiment polarity, and a categorical engagement type (e.g., *product inquiry*, *comparison*, *price inquiry*, *purchase intent*, *dismissal*). This contrasts with most prior LLM-as-judge deployments, which treat the judge as an offline evaluator of a held-out test set. Using the judge as an *in-the-loop* instrument raises new considerations—judge consistency across runs, the cost of an additional LLM call per engaged turn, and the risk of judge drift as the underlying model is updated—that are under-explored in the current literature and which this project confronts directly.

2.6 User Engagement Metrics in Conversational Systems

Traditional ad metrics. The standard measurement framework for digital advertising centres on three quantities: impressions (the number of times an ad is rendered), click-through rate (CTR, the fraction of impressions that result in a click), and conversion rate (the fraction of clicks that result in a desired action, such as a purchase). These metrics are well-understood and form the basis of pricing models including cost per mille (CPM) and cost per click (CPC). However, they were designed for a visual medium in which the user’s attention is inferred from the rendering of a display unit. In a conversational interface, where the entire response is a continuous text stream, the concept of a discrete “impression” is less well defined, and a “click” on an embedded link may not capture the full spectrum of user engagement with the recommended product.

A multi-dimensional engagement vector. In practice the Ad LLM platform decomposes “engagement” into a vector of six signals produced by a judge model on each post-ad follow-up turn: (a) a binary engaged/not-engaged flag; (b) an *engagement type* drawn from a fixed taxonomy (product inquiry, comparison, price inquiry, purchase intent, feature question, positive reaction, negative reaction, dismissal); (c) a continuous engagement score in $[0, 1]$; (d) an ad-naturalness score in $[0, 1]$ estimating how organic the placement felt from the user’s reaction; (e) a purchase-proximity score in $[0, 1]$ approximating how close the user appears to a buying decision; and (f) a sentiment polarity toward the ad. This decomposition is more informative than a single scalar engagement value because it separates *whether* the user engaged from *how* they engaged and *how the ad was received*, enabling the downstream optimisation described in Section 2.7 below. It also aligns with the direction advocated by Mehri and Eskénazi [27], who argue that dialogue quality is inherently multi-faceted and resists collapse to a single number.

Why conversational engagement is a stronger signal. The fundamental advantage of conversational engagement metrics over traditional click-through measurement is that they capture *active cognitive processing* rather than a binary click-or-not-click signal. A user who asks a follow-up question about an advertised product has read the recommendation, understood it, found it relevant, and invested the effort to formulate a new query. This chain of cognitive steps is substantially more informative than a click, which may be accidental, curiosity-driven, or immediately followed by a bounce. Conversational engagement therefore offers advertisers a higher-fidelity signal of genuine interest, which in turn supports more accurate attribution and, ultimately, more efficient allocation of advertising spend.

Related work on measuring user attention in dialogue. Research on dialogue systems has developed several proxies for user engagement, including turn length, response latency, sentiment trajectory, and topic persistence [27], [39]. See et al. [39] found that longer user responses and sustained topic continuity correlate with self-reported satisfaction in open-domain dialogue. Mehri and Eskénazi [27] proposed unsupervised dialogue quality metrics based on response coherence and relevance scoring. The engagement classifier in the Ad LLM platform draws on this work by treating topic persistence (continued discussion of the advertised product) as one component of engagement, but goes beyond lexical proxies by delegating the full classification to an LLM judge that returns the structured six-signal vector described above. This replaces an earlier keyword-and-entity-matching prototype, and is made tractable by the same LLM-as-judge infrastructure used for offline quality monitoring.

2.7 Adaptive Prompt and Context Optimisation

From static prompts to data-driven prompt refinement. The prompt-engineering literature surveyed in Section 2.5 treats the system prompt as a human-authored artefact that is iterated manually against an evaluation set. A growing body of work has begun to automate this loop. Automatic Prompt Engineer (APE) [49] frames prompt design as a black-box search problem in which candidate instructions are generated by an LLM, scored against a task metric, and resampled. PromptBreeder [15] extends this to a self-referential evolutionary process in which both the task prompts and the mutation operators are improved jointly. These approaches demonstrate that prompts are not fixed assets but can be optimised against a measurable reward signal—essentially applying the reinforcement-learning-from-feedback idea at the prompt level rather than the weight level.

Per-item context as an optimisable unit. The Ad LLM platform applies this principle at the granularity of individual ads. Each ad in the inventory carries, in addition to its creative text and targeting metadata, a free-text “presentation context guide” that is injected into the response-synthesis prompt whenever that ad is matched. This guide specifies how the product should be framed: tone, recommended angles, features to emphasise, situations in which to avoid mentioning it, and comparisons to avoid. It is distinct from the creative text in that it constrains the *model’s rendering* of the product rather than the copy shown to the user. The context guide is versioned: each update is persisted alongside the engagement metrics snapshot that justified it, yielding an auditable history of how the framing of each ad has evolved. The initial version of each context guide is itself generated automatically when the ad is first created: a bootstrap LLM call inspects the ad’s creative text, targeting metadata, and (where available)

scraped product-page content, and drafts a first-pass guide, so that the optimisation loop begins from a reasonable starting point rather than from an empty string.

Closing the loop with engagement feedback. The multi-dimensional engagement vector introduced above serves as the reward signal for an automatic optimiser. At a configurable cadence, and subject to minimum-sample and cooldown guards, a separate LLM call reviews the recent engagement history for an ad—aggregate engagement rate, average engagement score, average naturalness score, average purchase proximity, positive and negative sentiment counts, dismissal counts, and a sample of raw follow-up messages—together with the current context guide, and proposes a revised guide with an accompanying natural-language rationale. The optimiser prompt encodes diagnostic heuristics such as “if naturalness is low, soften the tone” and “if dismissals are high, tighten the relevance criteria.” The resulting system implements a closed loop analogous in spirit to bandit-style ad optimisation [26], but operating over natural-language prompt fragments rather than numeric bid or placement parameters, and using an LLM both to interpret the feedback and to author the next candidate. To the best of our knowledge, no prior published work applies per-item LLM-driven prompt optimisation to a conversational advertising setting.

Relationship to reinforcement learning from human feedback. RLHF [32] trains model weights against a reward model learned from human preference data. The adaptive context mechanism adopts the same underlying intuition—behaviour should be shaped by downstream feedback—but differs in two important ways. First, the feedback signal is produced by an LLM judge rather than human raters, following the LLM-as-judge paradigm discussed above. Second, no gradient updates are applied to the base model; adaptation happens entirely in the text of the prompt context, which is cheap, reversible, and interpretable. This positions the approach closer to *prompt-level learning* than to parameter-level fine-tuning, and makes it attractive for deployment scenarios in which the base model is a third-party API whose weights are inaccessible.

2.8 Generative Image Models and Native Product Placement

Text-to-image and image-editing models. Recent text-to-image systems—DALL-E 3 [5], Stable Diffusion [34], and Imagen [36]—produce photorealistic images from natural-language prompts and have become standard components of generative chat interfaces. A parallel line of work on image editing and subject-driven generation, including DreamBooth [35] and InstructPix2Pix [6], enables a specific visual subject to be inserted or preserved across generations by conditioning on one or more reference images in addition to the text prompt. These capabilities are now exposed as image-editing endpoints in commercial APIs, allowing a reference image to act as a visual anchor for the generated scene.

Product placement as a creative convention. In traditional media—film, television, and more recently influencer video—product placement is a long-established advertising format in which a branded item is integrated diegetically into the scene rather than presented as an interruption [4]. Empirical studies have shown that well-executed placement can achieve higher recall and more favourable brand attitudes than conventional spot advertising, provided the placement feels natural to the narrative context [33]. Until recently, this format required human

creative direction and physical staging; generative image models now make algorithmic placement technically feasible at per-request granularity.

Extending the Ad LLM thesis to the visual modality. When a user issues an image-generation request through the Ad LLM platform, the same context-extraction and RAG-based ad-matching pipeline that handles chat messages is invoked to select a relevant product. An auxiliary LLM call then rewrites the user’s original image prompt so that the matched product appears organically in the scene (e.g., “on a shelf”, “held by a subject”, “worn by a runner”), preserving the user’s original creative intent while adding the product as a diegetic element. If the ad inventory includes a reference photograph of the product, the rewritten prompt is passed to an image-editing endpoint together with the reference image, so that the generated scene is visually grounded in the actual product rather than in the model’s prior over product names. This directly mirrors the text pipeline’s “instruct the generative model to weave in the ad” design philosophy and extends it to pixels, producing what is, to our knowledge, the first described pipeline for *RAG-matched, sponsored product placement in user-requested AI-generated imagery*.

Open questions specific to the visual modality. Placing real branded products in generated images raises ethical and legal questions that do not arise in the text case. Visual disclosure is harder than textual disclosure: a [Sponsored] tag can be prepended to a response caption, but the image itself carries no intrinsic marker that the product within it is a paid placement. Intellectual-property considerations are also sharper, since a generated image of a branded product may implicate trademark and trade-dress rights that a textual mention does not. These issues are acknowledged as open and are discussed further in the ethical considerations of Section ??; the literature on regulatory responses to AI-generated visual content is still nascent and, at the time of writing, does not yet provide clear guidance on sponsored generative imagery.

2.9 Summary and Research Gap

The preceding subsections have surveyed eight bodies of work that converge on the problem addressed by this project: (i) the evolution of digital advertising toward native, content-integrated formats; (ii) the capabilities of transformer-based LLMs for instruction following, structured output, and tone-adapted generation; (iii) the RAG paradigm for grounding language-model output in retrieved external context; (iv) the emerging practice of embedding advertising in conversational AI interfaces, together with the ethical and regulatory constraints that govern disclosure; (v) prompt engineering techniques for controlled generation and the LLM-as-judge paradigm for both offline and in-the-loop quality evaluation; (vi) engagement metrics that go beyond click-through to capture genuine user attention in dialogue systems, decomposed into a multi-dimensional signal vector; (vii) automatic and data-driven prompt optimisation as a lightweight alternative to parameter-level fine-tuning; and (viii) generative image models and the long-standing advertising convention of product placement, which together make algorithmic visual ad integration newly feasible.

The gap that this project addresses lies at the intersection of these strands. Existing industry deployments of conversational advertising—Microsoft’s Bing Chat ads and Google’s SGE/AI Mode placements—insert conventional ad units (text links, product cards) alongside AI-generated responses rather than instructing the language model to integrate product mentions into the

response itself. Academic work on native advertising has extensively studied user perception and disclosure effectiveness, but has not examined LLM-generated native advertising in which the model itself produces the sponsored content. The RAG literature provides the retrieval infrastructure for context-grounded generation, but has not applied the retrieve-then-generate pipeline to ad matching and integration. Prompt engineering research offers the techniques for controlled output, but has not specifically addressed the dual-objective problem of simultaneous helpfulness and natural ad integration.

No open, well-documented architecture currently exists that combines semantic ad matching via a specialised RAG pipeline, LLM-driven natural ad integration with mandatory disclosure, sensitivity-aware ad gating, and engagement-based measurement within a single end-to-end system. The *Ad LLM as a Service* platform is designed to address this gap: it provides a complete, working pipeline from user message to ad-augmented response, accompanied by an analytics framework that measures conversational engagement as a first-class advertising metric.

3 Problem Statement

3.1 Limitations of Traditional Ad Formats in Conversational AI

3.2 Technical Challenges

3.2.1 Real-Time Intent Extraction from Multi-Turn Conversations

3.2.2 Semantic Ad Matching at Low Latency

3.2.3 Natural Ad Integration Without Degrading User Experience

3.2.4 Multi-Dimensional Engagement Measurement

3.2.5 Closing the Feedback Loop: Adaptive Ad Presentation

3.2.6 Campaign and Budget Management at Request Time

3.2.7 Sponsored Visual Generation

3.3 Ethical and Regulatory Considerations

3.4 Formal Problem Definition

4 Solution Design

4.1 System Architecture Overview

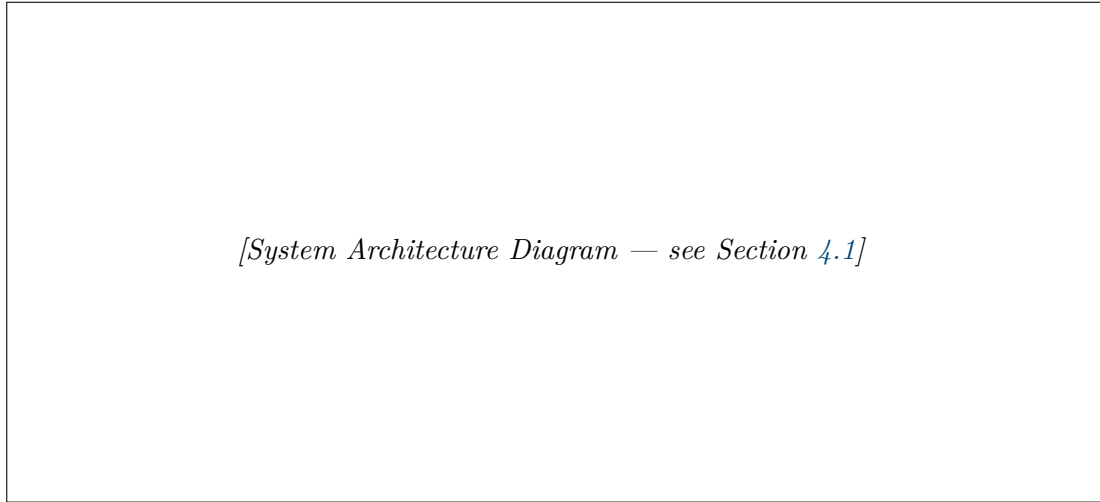


Figure 1. High-level system architecture of the Ad LLM platform, including the synchronous request path (chat and image) and the asynchronous engagement-to-context feedback loop.

Table 1. Technology stack summary.

Component	Technology
Chat Frontend	React
Analytics & Portal UI	React + Recharts
Orchestrator API	FastAPI (Python)
Session Store	Redis
Relational + Vector DB	PostgreSQL + pgvector
Embedding Model	text-embedding-3-small (OpenAI)
Context Extraction LLM	Groq (lightweight)
Response Generation LLM	Groq (capable)
Engagement Judge LLM	Groq (JSON-mode)
Context Optimiser LLM	Groq
Image Generation Model	OpenAI gpt-image-1 (generate / edit)
Event Logging	PostgreSQL (JSONB payload)

4.2 Component Design

4.2.1 Orchestrator API (FastAPI)

4.2.2 Context Extraction Module

4.2.3 Session Management (Redis)

4.2.4 Ad Inventory, Campaigns, and Advertisers (PostgreSQL + pgvector)

4.2.5 Embedding and Semantic Matching Pipeline

4.2.6 Re-Ranking, Budget Gating, and Business Logic

Algorithm 1 Ad Candidate Re-Ranking with Budget Gating

Input: Query embedding $\mathbf{q}$, context C , session ID s

Output: Ranked list of top- k ad candidates

- 1: $raw \leftarrow \text{VectorSearch}(\mathbf{q}, \text{limit} = 20)$ $\triangleright$ pgvector cosine similarity
 - 2: $shown \leftarrow \text{Redis.get}(s).\text{ads_shown}$
 - 3: $eligible \leftarrow \{a \in raw \mid a.\text{active} \wedge a.\text{id} \notin shown \wedge \text{CHECKBUDGET}(a.\text{campaign_id})\}$ $\triangleright$ budget / schedule gate
 - 4: $b_{\max} \leftarrow \max_{a \in eligible} a.\text{bid_cpm}$
 - 5: **for all** $a \in eligible$ **do**
 - 6: $a.\text{score} \leftarrow 0.5 \cdot a.\text{sim} + 0.3 \cdot \frac{a.\text{bid_cpm}}{b_{\max}} + 0.2 \cdot C.\text{purchase_signal}$
 - 7: **end for**
 - 8: **return** $\text{TopK}(eligible, k = 3)$
-

*4.2.7 Response Synthesis with Ad Injection**4.2.8 Image Generation with Product Placement**4.2.9 LLM-Judge-Based Engagement Detection**4.2.10 Adaptive Ad Context Optimisation**4.2.11 Click Tracking, Event Logging, and Budget Accounting**4.2.12 Advertiser Portal and Campaign Management**4.2.13 Analytics API and Dashboards***4.3 Interface Contracts and Workstream Division****Table 2.** Interface contracts between Backend and AI workstreams.

Function	Caller → Callee	Description
<code>extract_context</code>	Backend → AI	Structured context from user message and history
<code>embed_query</code>	Backend → AI	Query embedding from context
<code>embed_ad</code>	Backend → AI	Ad embedding at ingestion time
<code>get_candidate_ads</code>	AI → Backend	Ranked candidates via vector search + budget gating
<code>generate_response</code>	Backend → AI	Text response with optional ad injection and context guide
<code>generate_image</code>	Backend → AI	Image with product placement
<code>detect_engagement</code>	Backend → AI	Multi-dimensional engagement vector from follow-up turn
<code>generate_initial_context</code>	Backend → AI	Seed presentation guide on ad creation
<code>optimize_ad_context</code>	Backend (self)	Revise presentation guide from recent engagement metrics
<code>extract_product_from_url</code>	Backend → AI	Ad-ready fields from scraped product page

4.4 Quality Assurance and Evaluation Framework**4.5 Error Handling and Graceful Degradation****5 Illustrative Examples**

5.1 Example 1: Product Research Conversation**5.2 Example 2: Sensitive Conversation (Ad Suppression)****5.3 Example 3: Irrelevant Ad Rejection****5.4 Example 4: Multi-Dimensional Engagement Follow-Up****5.5 Example 5: Adaptive Context Optimisation Loop****5.6 Example 6: Image Generation with Product Placement****5.7 Example 7: Campaign Lifecycle and Budget Exhaustion****6 Conclusions**

6.1 Summary of Contributions**6.2 Workstream Contributions****6.3 Limitations****6.4 Future Work****6.5 Reflections**

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A Database Schema

B API Endpoint Specifications

C Prompt Templates

D Context Extraction Test Cases

E Response Synthesis Evaluation Set

F Session State Schema

G Project Timeline

Table 3. Project timeline by workstream.

Week	Backend Workstream	AI/LLM Workstream	Milestone
1	Orchestrator skeleton, Redis, Postgres + pgvector schema	Context extraction prompt design + module	Infrastructure foundation
2	Ad CRUD API, bulk ingestion, re-ranking logic	Embedding functions, retrieval quality testing	Ad matching pipeline ready
3	Click tracking, event logging, tracking link generator	Response synthesis prompt engineering + module	Core pipeline complete
4	Analytics API, frontend integration, end-to-end wiring	Eval set, quality tuning, engagement classifier	End-to-end integration
5	Error handling, monitoring, hardening	LLM-as-judge evaluator, attention model start	Demo-ready MVP

